

# WEEK-I

Figure 1: Creation and selection of the BookFlow database

```
root@mrudula:~# mysql -u root
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 10
Server version: 9.7.1 Homebrew

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owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> CREATE DATABASE IF NOT EXISTS bookflow_db;
Query OK, 1 row affected, 1 warning (0.06 sec)

mysql> USE bookflow_db;
Reading table information for completion of table and column names
You can turn off this feature to get a quicker startup with -A
Database changed
mysql>
```

Figure 2: Displaying the tables available in the BookFlow database

```
[mysql> SHOW TABLES;
+-----+
| Tables_in_bookflow_db |
+-----+
| Books                  |
| Donation_History       |
| Loans                  |
| Members                |
+-----+
4 rows in set (0.001 sec)

mysql>
```

Figure 3: Verifying the Books table structure and constraints

```
[mysql> DESCRIBE Books;
+-----+-----+-----+-----+-----+-----+
| Field          | Type          | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| book_id        | int           | NO   | PRI | NULL    |       |
| title          | varchar(100)  | NO   |     | NULL    |       |
| isbn           | varchar(20)   | YES  | UNI | NULL    |       |
| published_year | int           | YES  |     | NULL    |       |
+-----+-----+-----+-----+-----+-----+
4 rows in set (0.002 sec)

mysql>
```

Figure 4: Verifying the Members table structure and constraints

```
[mysql> DESCRIBE Members;
+-----+-----+-----+-----+-----+-----+
| Field      | Type          | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| member_id  | int           | NO   | PRI | NULL     |       |
| full_name  | varchar(100)  | YES  |     | NULL     |       |
| email      | varchar(100)  | YES  | UNI | NULL     |       |
+-----+-----+-----+-----+-----+-----+
3 rows in set (0.002 sec)

mysql> 
```

Figure 5: Displaying the Books records in the BookFlow database

```
[mysql> SELECT * FROM Books;
+-----+-----+-----+-----+
| book_id | title          | isbn   | published_year |
+-----+-----+-----+-----+
| 1       | Python Programming | ISBN001 | 2022           |
| 2       | Database Systems  | ISBN002 | 2021           |
| 3       | Machine Learning  | ISBN003 | 2024           |
| 4       | Java Programming  | ISBN004 | 2025           |
+-----+-----+-----+-----+
4 rows in set (0.000 sec)

mysql> 
```

Figure 6: Displaying the Members records in the BookFlow database

```
[mysql> SELECT * FROM Members;
+-----+-----+-----+
| member_id | full_name | email          |
+-----+-----+-----+
| 101       | Amrutha   | amrutha@gmail.com |
| 102       | Rahul     | rahul@gmail.com   |
| 103       | Priya     | priya@gmail.com   |
+-----+-----+-----+
3 rows in set (0.000 sec)

mysql> 
```

Figure 7: UNIQUE constraint preventing duplicate ISBN insertion

```
mysql> INSERT INTO Books (title, isbn, published_year)
VALUES ('Fake copy', '9780061122415% 2000) ;
ERROR 1364 (HY00): Field 'book_id' doesn't have a default value
mysql> I
```

Figure 8: NOT NULL constraint preventing insertion of a book without a title

```
mysql> INSERT INTO Books (title, isbn, published_year)
-> VALUES (NULL, '99999999999999% 2010); ERROR
1048 (23000): Column 'title' cannot be null mysql> I
```

Figure 9: CHECK constraint preventing a future publication year

```
mysql> INSERT INTO Books (title, isbn, published_year)
      -> VALUES ( 'Time Traveler' , '88888888888888%
ERROR 1364 (Hyeeo): Field 'book_id' doesn't have a default value
mysql> I
```

Figure 10: UNIQUE constraint preventing duplicate member email

```
mysql> INSERT INTO Members (full_name, email)
      -> VALUES ( 'Anil Clone', anil.kumar@example.com ' ');
ERROR 1364 (HYeee) : Field 'member_id' doesn't have a default value
mysql> I
```

INTO

Figure 11: Final verification of Books and Members records and table constraints

```
mysql> SELECT * FROM Books;
+----+-----+-----+
| book_id | title                | isbn      | published_year |
+----+-----+-----+
| 1 | Python Programing    | ISBN0001  | 2022           |
| 2 | Database Systems     | ISBN0002  | 2021           |
| 3 | Machine Learning     | ISBN0003  | 2024           |
| 4 | Java Programing      | ISBN0004  | 2025           |
+----+-----+-----+
4 rows in set (0.00e sec)

mysql> SELECT * FROM Members;
+----+-----+-----+
| member_id | full_name | email                |
+----+-----+-----+
| 101 | Amrutha | amrutha@gmail.com |
| 102 | Rahul   | rahul@gmail.com    |
| 103 | Priya   | priya@gmail.com    |
+----+-----+-----+
3 rows in set (0.00e sec)

mysql> DESCRIBE Books;
+----+-----+-----+-----+-----+-----+
| Field | Type | Null | Key | Default | Extra |
+----+-----+-----+-----+-----+-----+
| book_id | int | NO | PRI | NULL | |
| title | varchar(100) | NO | | NULL | |
| isbn | varchar(20) | YES | | NULL | |
| published_year | int | YES | | NULL | |
+----+-----+-----+-----+-----+-----+
4 rows in set (0.00e sec)

mysql> DESCRIBE Members;
+----+-----+-----+-----+-----+-----+
| Field | Type | Null | Key | Default | Extra |
+----+-----+-----+-----+-----+-----+
| member_id | int | NO | PRI | NULL | |
| full_name | varchar(100) | YES | | NULL | |
| email | varchar(100) | YES | NI | NULL | |
+----+-----+-----+-----+-----+-----+
3 rows in set (0.002 sec)

mysql> I
```